1. Introduction

The telecom industry faces an average annual churn rate of 30-35%, While acquiring new customers costs 5-10 times more than retaining existing ones. (1)

Customer churn is a key indicator of a company's health impacting customer lifetime value (CLV), brand loyalty, and revenue stability. For the competitive telecoms industry, reducing customer churn is key to maintaining market share.

1.1 The objective of this research

1. Identify customers who are likely to churn in the future.
2. Identify key factors affecting customer churn and analyze which factors increase the risk of churn.
3. Provide data-driven insights to formulate customer retention strategy and reduce customer churn rate.

2. Data Overview

This data set comprises 7,043 customers' data from a telecommunication service provider with 21 attributes (columns) including "churn" status. Regarding the "Dtype" column on the table on the right, there are 18 attributes being objects (either a string or a mix of other data types), 2 attributes being integers, and 1 attribute being float.

From this overview, "SeniorCitizen" and "TotalCharges" are different from other columns in a similar fashion. "SeniorCitizen" can either be in string format (Yes and No) or integer format (Yes = 1 and No = 0). However, the data type in "TotalCharges" must turn into a floating-point number before making further studies on this dataset.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
```

Figure 1. Original Data Information

	Column Name	Column Meaning
1	customerID	ID of individual customer
2	gender	Gender of customer
3	SeniorCitizen	Customer that is senior citizen
4	Partner	Similar to mariage status
5	Dependents	Customer that have dependents
6	tenure	Duration of using service (Month)
7	PhoneService	Using phone service
8	MultipleLines	Using both phone service and internet service
9	InternetService	Type of internet service used by customer
10	OnlineSecurity	Using online security
11	OnlineBackup	Using online backup
12	DeviceProtection	Using device protection
13	TechSupport	Using tech support
14	StreamingTV	Using TV streaming
15	StreamingMovies	Using movies streaming
16	Contract	Term of contract (1 month, 1 year, 2 years)
17	PaperlessBilling	Using paperless billing
18	PaymentMethod	Payment method that customer use
19	MonthlyCharges	Monthly service fee
20	TotalCharges	Total charges since begin using the service
21	Churn	Customer churn status

Figure 2. Meaning of each attribute

3. Data Preprocessing

In this process, the main objective is to provide datasets that can be used to visualize and build prediction models for the smoothness of descriptive analysis and the machine learning process.

```
[11] missing_total_charges_count = overview['TotalCharges'].str.count(' ').sum()
     print(missing_total_charges_count)

     11
```

Figure 3. Method of Finding Missing Value in “TotalCharges”

First, we identified and searched for string and null values in the “TotalCharges” column. After several methods, 11 values in this column were found to be empty strings as shown in the above

figure. After that, we used the replace function in Python to replace those empty strings with 0. As a result, all of the values in the "TotalCharges" become floating-point numbers.

Second, we developed three versions of the dataset for implementation for different purposes. Generating new columns with new interpretations, applying numerical encoding, and eliminating unutilized data were examples of methods for constructing new datasets.

Version 1	used for the visualization process through Excel.
Version 2	used for the correlation analysis which required data in binary format.
Version 3	built regarding requirements of data scientists for machine learning.

Version 1

```
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   gender             7043 non-null   object
 1   Partner            7043 non-null   object
 2   Dependents         7043 non-null   object
 3   tenure             7043 non-null   int64
 4   PhoneService       7043 non-null   object
 5   InternetService    7043 non-null   object
 6   contract_month     7043 non-null   int64
 7   PaperlessBilling   7043 non-null   object
 8   auto_bill          7043 non-null   object
 9   MonthlyCharges     7043 non-null   float64
 10  Churn              7043 non-null   int64
 11  senior_yes         7043 non-null   object
 12  multiple_lines     7043 non-null   object
 13  online_security    7043 non-null   object
 14  online_backup      7043 non-null   object
 15  device_protection  7043 non-null   object
 16  tech_support       7043 non-null   object
 17  streaming_tv       7043 non-null   object
 18  streaming_movies   7043 non-null   object
 19  payment_method     7043 non-null   object
 20  total_charges      7043 non-null   float64
dtypes: float64(2), int64(3), object(16)
```

Version 2

```
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 34 columns):
 #   Column                                 Non-Null Count  Dtype
---  ------                                 --------------  -----
 0   tenure                                 7043 non-null   int64
 1   MonthlyCharges                         7043 non-null   float64
 2   Churn                                  7043 non-null   int64
 3   total_charges                          7043 non-null   float64
 4   Partner_Yes                            7043 non-null   int64
 5   Dependents_Yes                         7043 non-null   int64
 6   PhoneService_Yes                       7043 non-null   int64
 7   MultipleLines_No phone service         7043 non-null   int64
 8   MultipleLines_Yes                      7043 non-null   int64
 9   OnlineSecurity_No internet service     7043 non-null   int64
 10  OnlineSecurity_Yes                     7043 non-null   int64
 11  DeviceProtection_No internet service   7043 non-null   int64
 12  DeviceProtection_Yes                   7043 non-null   int64
 13  TechSupport_No internet service        7043 non-null   int64
 14  TechSupport_Yes                        7043 non-null   int64
 15  StreamingTV_No internet service        7043 non-null   int64
 16  StreamingTV_Yes                        7043 non-null   int64
 17  StreamingMovies_No internet service    7043 non-null   int64
 18  StreamingMovies_Yes                    7043 non-null   int64
 19  PaperlessBilling_Yes                   7043 non-null   int64
 20  auto_bill_Yes                          7043 non-null   int64
 21  senior_Yes                             7043 non-null   int64
 22  gender_Female                          7043 non-null   int64
 23  gender_Male                            7043 non-null   int64
 24  InternetService_DSL                    7043 non-null   int64
 25  InternetService_Fiber optic            7043 non-null   int64
 26  InternetService_No                     7043 non-null   int64
 27  Contract_Month-to-month                7043 non-null   int64
 28  Contract_One year                      7043 non-null   int64
 29  Contract_Two year                      7043 non-null   int64
 30  payment_method_Bank transfer           7043 non-null   int64
 31  payment_method_Credit card             7043 non-null   int64
 32  payment_method_Electronic check        7043 non-null   int64
 33  payment_method_Mailed check            7043 non-null   int64
dtypes: float64(2), int64(32)
```

Version 3

```
Index: 7032 entries, 0 to 7042
Data columns (total 18 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   Contract                        7032 non-null   int64
 1   InternetService                 7032 non-null   object
 2   StreamingMovies                 7032 non-null   object
 3   billing_convenience             7032 non-null   int64
 4   OnlineSecurity                  7032 non-null   object
 5   MonthlyCharges                  7032 non-null   float64
 6   TotalCharges                    7032 non-null   float64
 7   Dependents                      7032 non-null   object
 8   PaymentMethod                   7032 non-null   object
 9   renewed_status                  7032 non-null   int64
 10  TechSupport                     7032 non-null   object
 11  near_contract_end               7032 non-null   object
 12  PaperlessBilling                7032 non-null   object
 13  Partner                         7032 non-null   object
 14  tenure                          7032 non-null   int64
 15  SeniorCitizen                   7032 non-null   int64
 16  Churn                           7032 non-null   object
 17  internet_streaming_interaction  7032 non-null   int64
dtypes: float64(2), int64(6), object(10)
```

Figure 4. Info of Three Datasets

The challenge faced in the data preprocessing process was how flexible data can be used to analyze and interpret through several methods.

4. Analysis & Data Visualization

The present analysis of telecommunications customer churn analyzed multiple demographic, account, and service-related characteristics to ascertain the primary determinants of customer attrition. From figure 5, there are 73.5% of remaining customers and 26.5% of churn customers. To get deeper insight, we conducted a correlation study to assess the influence of each variable on churn likelihood, along with visualizations to illustrate patterns.

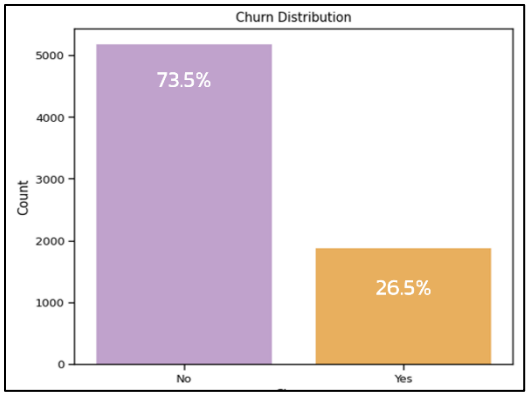


Figure 5. Distribution of Churn

Descriptive Stats	Tenure	MonthlyCharges	TotalCharges
Mean	32.42	64.8	2283.3
Standard Error	0.29	0.36	27.03
Median	29	70.35	1397.48
Mode	1	20.05	20.2
Standard Deviation	24.55	30.09	2266.77
Sample Variance	602.47	905.17	5138252.41
Kurtosis	-1.39	-1.26	-0.23
Skewness	0.24	-0.22	0.96
Range	71	100.5	8666
Minimum	1	18.25	18.8
Maximum	72	118.75	8684.8

Figure 6. Descriptive statistics on numerical data

4.1 Key Insights from Correlation Analysis

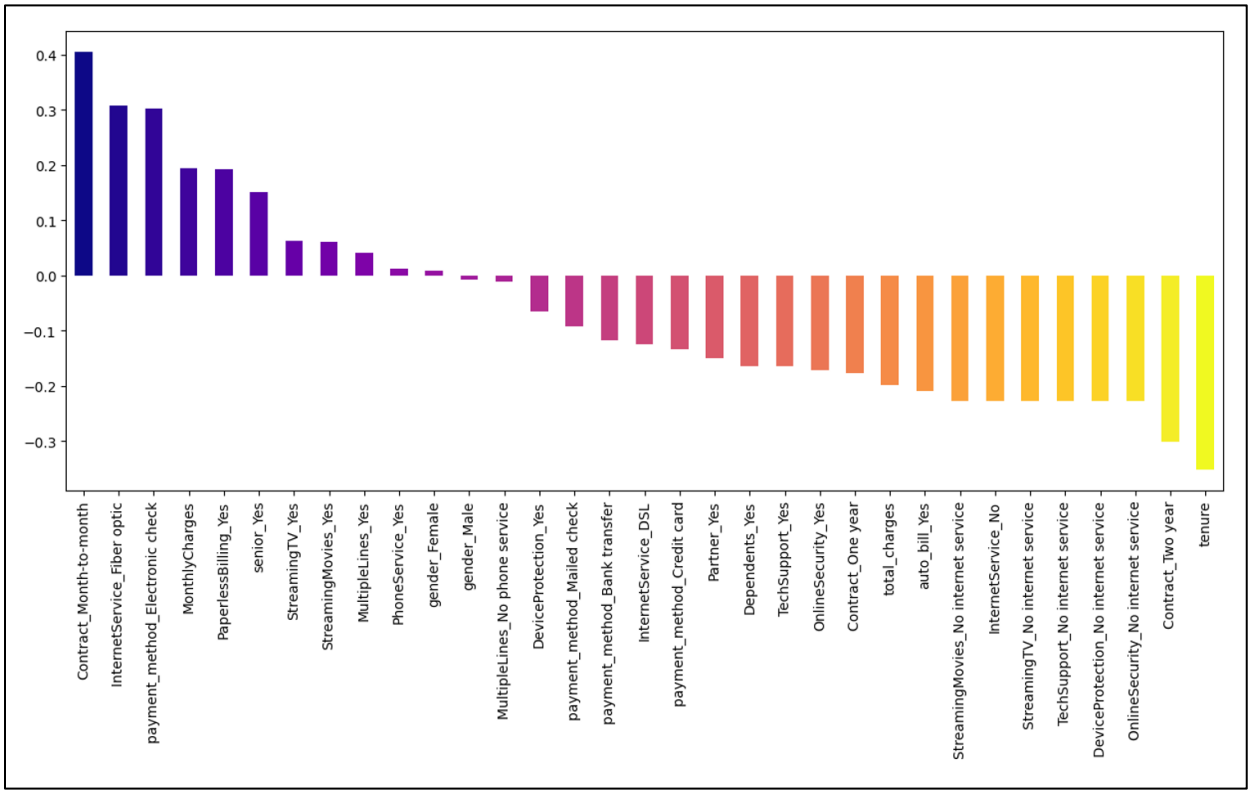


Figure 7. Correlation toward Churn

The correlation with churn was most significant for month-to-month contracts (0.41), indicating that shorter contracts heighten the probability of churn. On the other hand, one-year and two-year contracts exhibited negative associations (-0.18 and -0.30, each), indicating that extended commitments enhance retention.

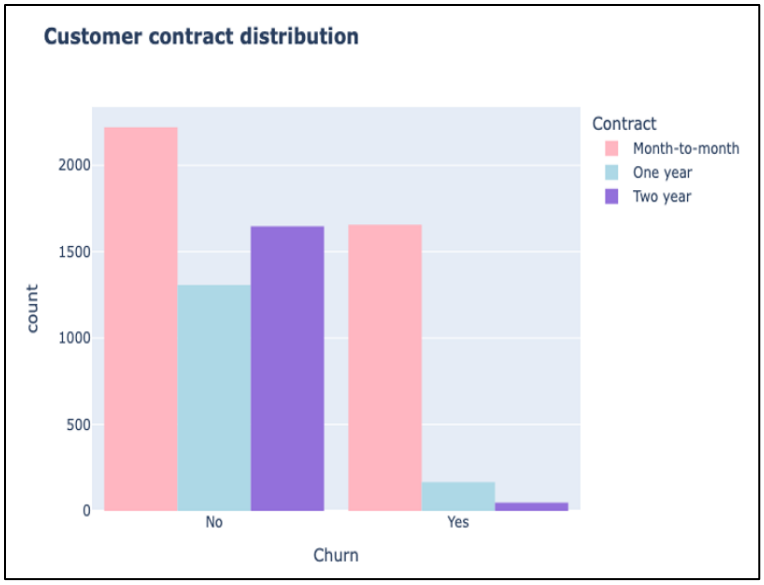


Figure 8. Customer Contract Distribution

Internet Service: Fibre optic service exhibited a positive correlation with churn (0.31), suggesting that fibre consumers experience greater churn rates, either attributable to competition or price sensitivity.

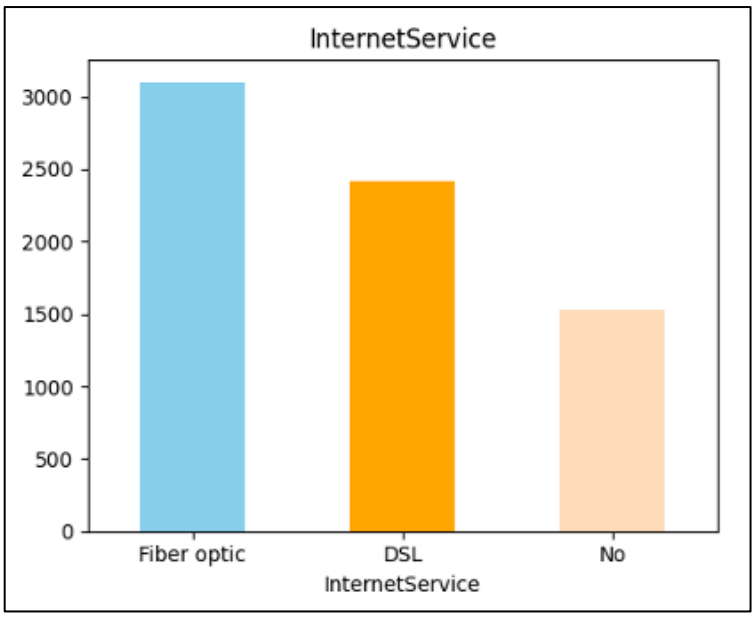


Figure 9. Internet Service

Payment Method: While bank transfers and credit cards showed a negative association with turnover (0.30), electronic cheque payments showed a positive correlation, indicating that the easy nature of payment may have an impact on retention.

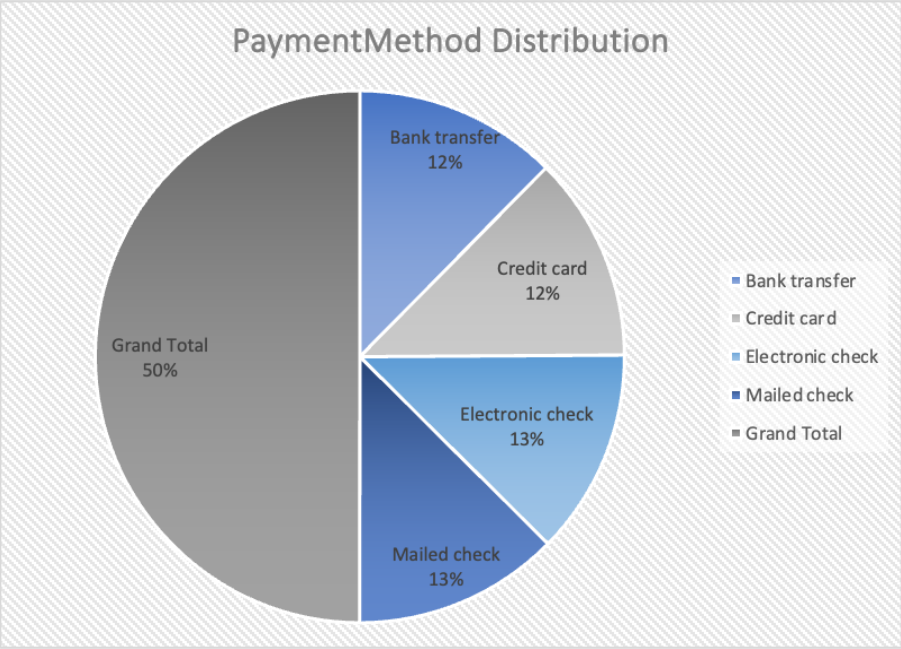


Figure 10. Payment Method Distribution

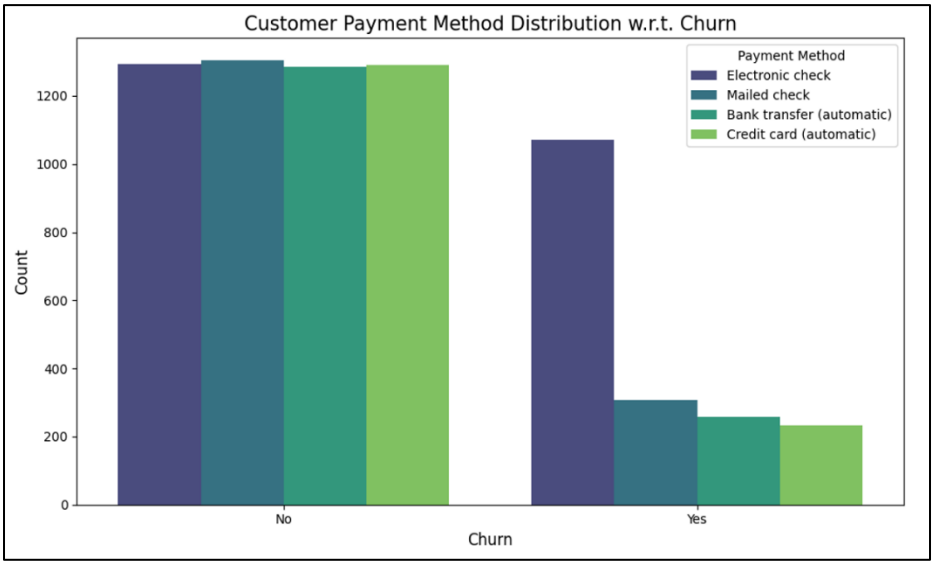


Figure 11. Customer Payment Method Distribution

4.2 Supplementary Visual Insights

Demographic Factors: The status of senior citizens had a weak positive connection (0.15) with churn, which is consistent with the visualization evidence showing that turnover is slightly more common among seniors. Customers with dependents exhibited a weak negative correlation (-0.16), indicating that dependents may enhance loyalty.

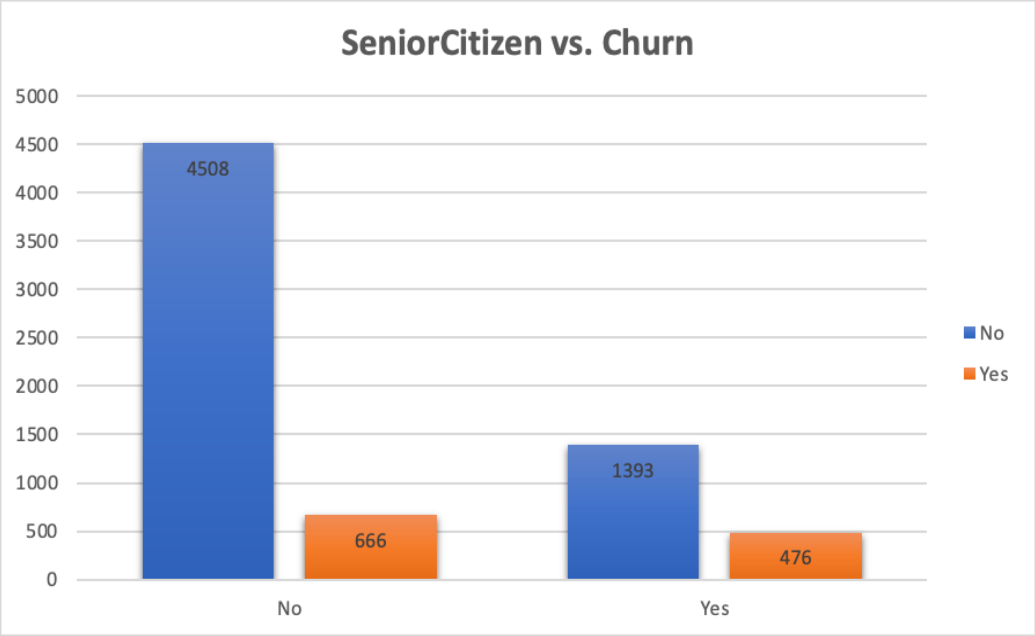


Figure 12. Senior Citizen by Churn

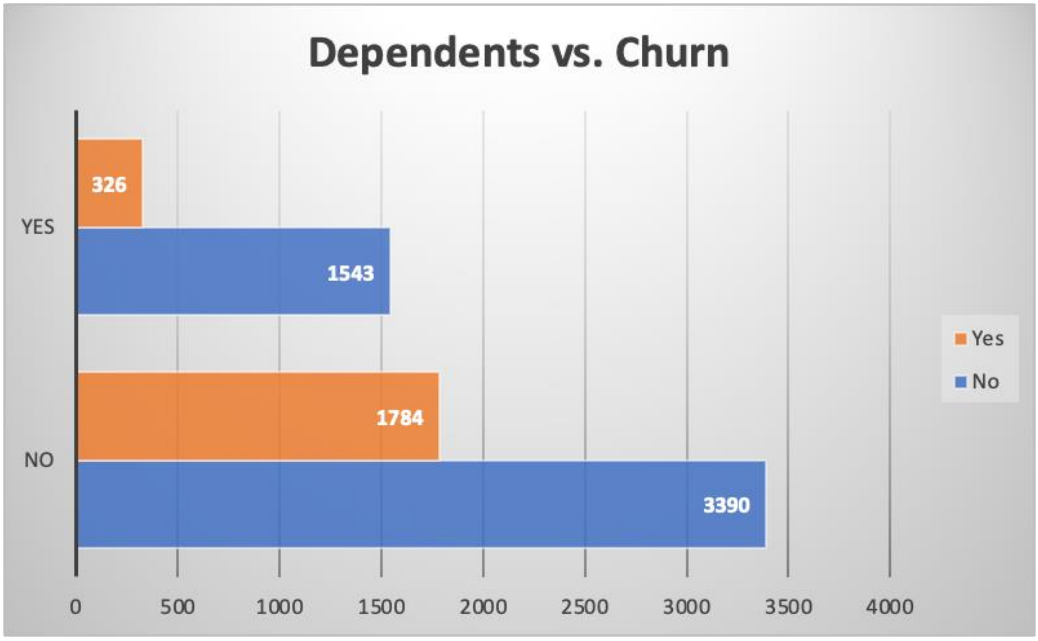


Figure 13. Dependents by Churn

Service Features: Variables such as Tech Support and Online Security exhibited moderate negative associations (-0.28 and -0.17), and graphic representations supported that customers utilizing these services experience lower turnover rates. These data indicate that additional services can mitigate turnover by increasing perceived value.

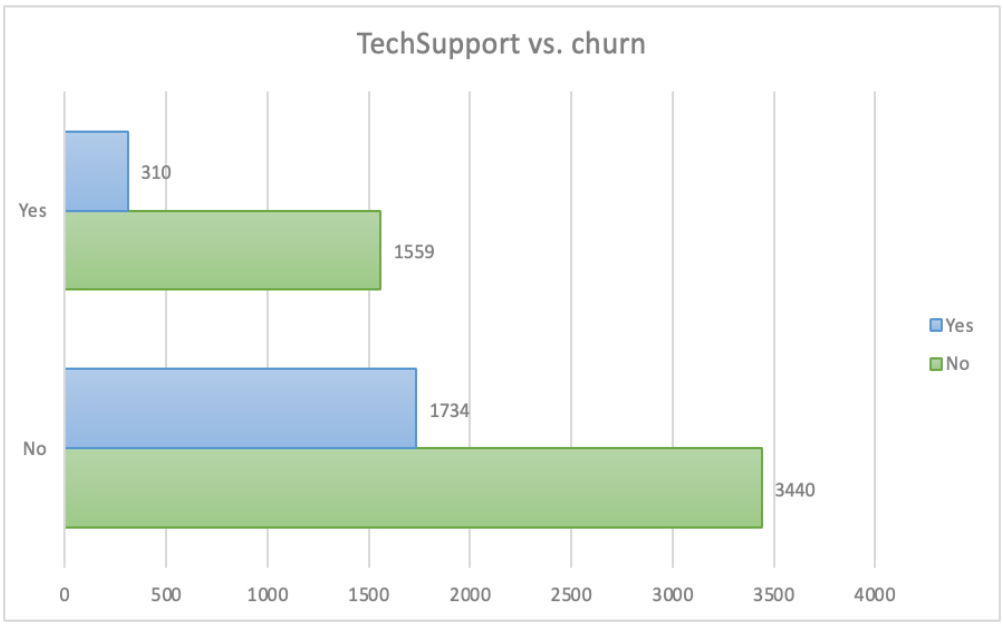


Figure 14. Tech Support in Churn

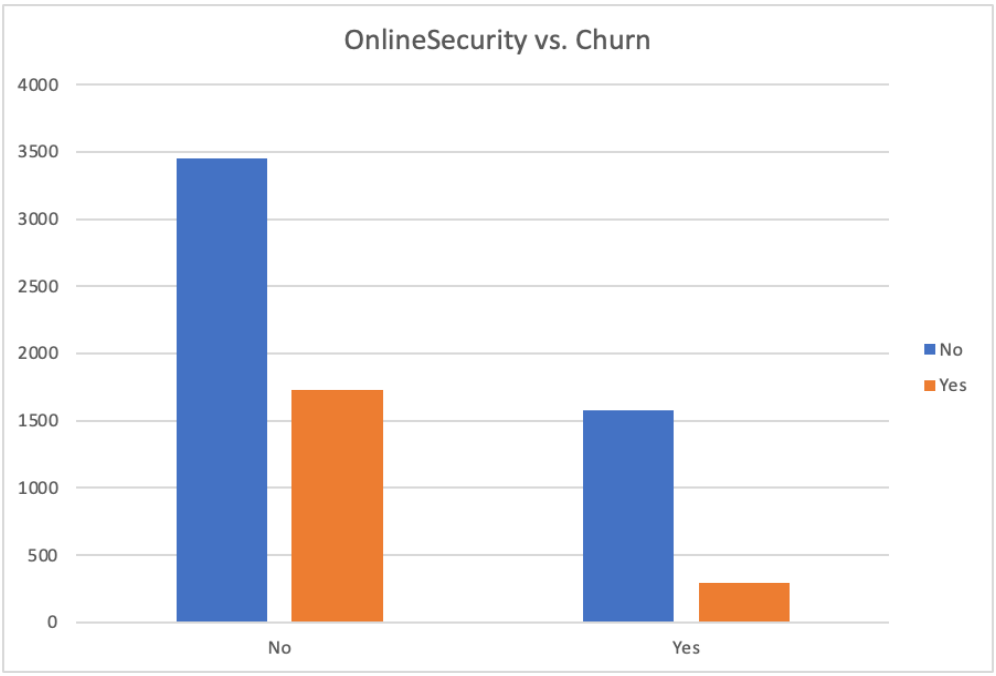


Figure 15. Online Security in Churn

The financial analysis revealed a weak positive correlation (0.19) between monthly charges and churn, suggesting that elevated monthly expenses may heighten the likelihood of customer attrition. Total charges had a minor negative correlation (-0.20), indicating that long-term customers may accept elevated expenses provided they perceive value.

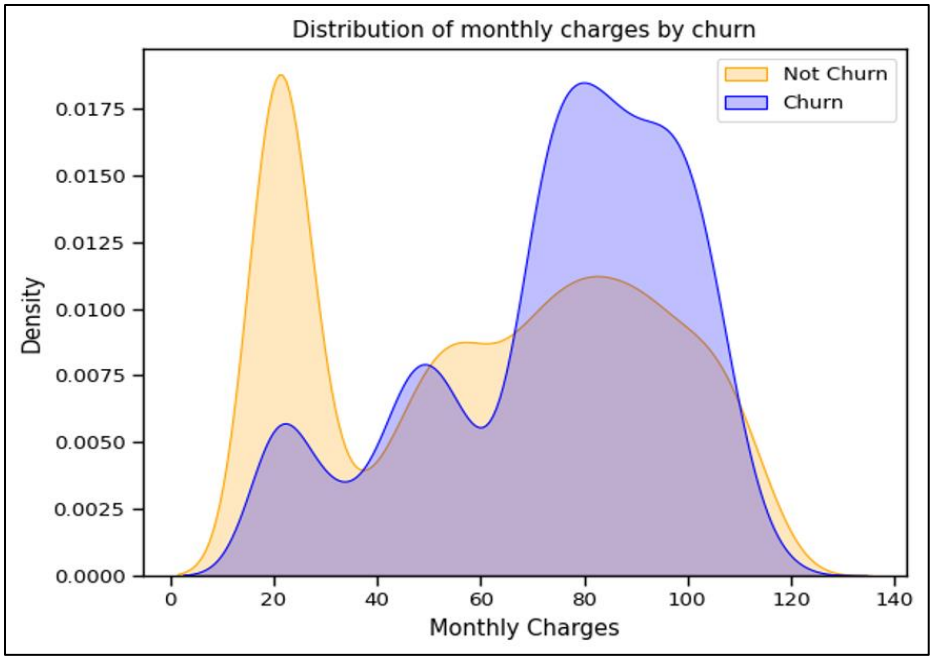


Figure 16. Distribution of monthly charges by churn

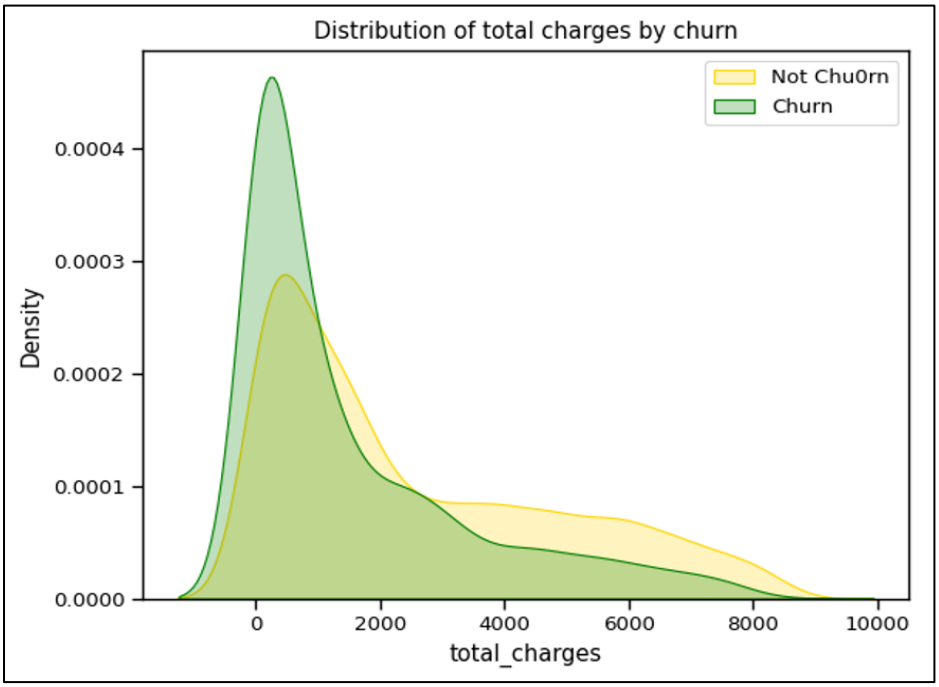


Figure 17. Distribution of the total charges by churn

4.3 Challenge

It was difficult to integrate correlation data with continuous and categorical variables, particularly when it came to making complicated multi-variable visualizations clear.

5. Feature Engineering

5.1 Label Encoding

We coded the labels of the non-numeric features in the version-3 dataset and converted them into numeric form.

```
OnlineSecurity : [0 2 1] = ['No' 'Yes' 'No internet service']
TechSupport : [0 2 1] = ['No' 'Yes' 'No internet service']
InternetService : [0 1 2] = ['DSL' 'Fiber optic' 'No']
Churn : [0 1] = ['No' 'Yes']
near_contract_end : [0 1] = ['no' 'yes']
Dependents : [0 1] = ['No' 'Yes']
StreamingMovies : [0 2 1] = ['No' 'Yes' 'No internet service']
Partner : [1 0] = ['Yes' 'No']
PaymentMethod : [2 3 0 1] = ['Electronic check' 'Mailed check' 'Bank transfer (automatic)'
 'Credit card (automatic)']
PaperlessBilling : [1 0] = ['Yes' 'No']
```

Figure 18. Results of label coding

5.2 Numerical Normalization

We use Min-Max Scaling to compress the values of each feature into a specified range, typically [0, 1].

renewed_status	StreamingMovies	Partner	TotalCharges	PaymentMethod	PaperlessBilling
0.244681	0.0	1.0	0.001275	0.666667	1.0
0.478723	0.0	0.0	0.215867	1.000000	0.0
0.255319	0.0	0.0	0.010310	1.000000	1.0
0.595745	0.0	0.0	0.210241	0.000000	0.0
0.255319	0.0	0.0	0.015330	0.666667	1.0

Figure 19. Results of numerical normalization

5.3 SMOTE (Synthetic Minority Over-sampling Technique)

SMOTE is a commonly used oversampling technique mainly used to deal with category imbalance in classification problems. In our analysis, we use SMOTE to address the imbalance in the number of Yes and No in the variable Churn.

```
Counter({0.0: 5163, 1.0: 5163})
```

Figure 20. Results of SMOTE

5.4 RFE (Recursive Feature Elimination)

RFE is a feature selection method that aims to improve the performance of a model by gradually eliminating unimportant features. In our analysis, we use Random Forest as the basic classifier for feature selection and model training.

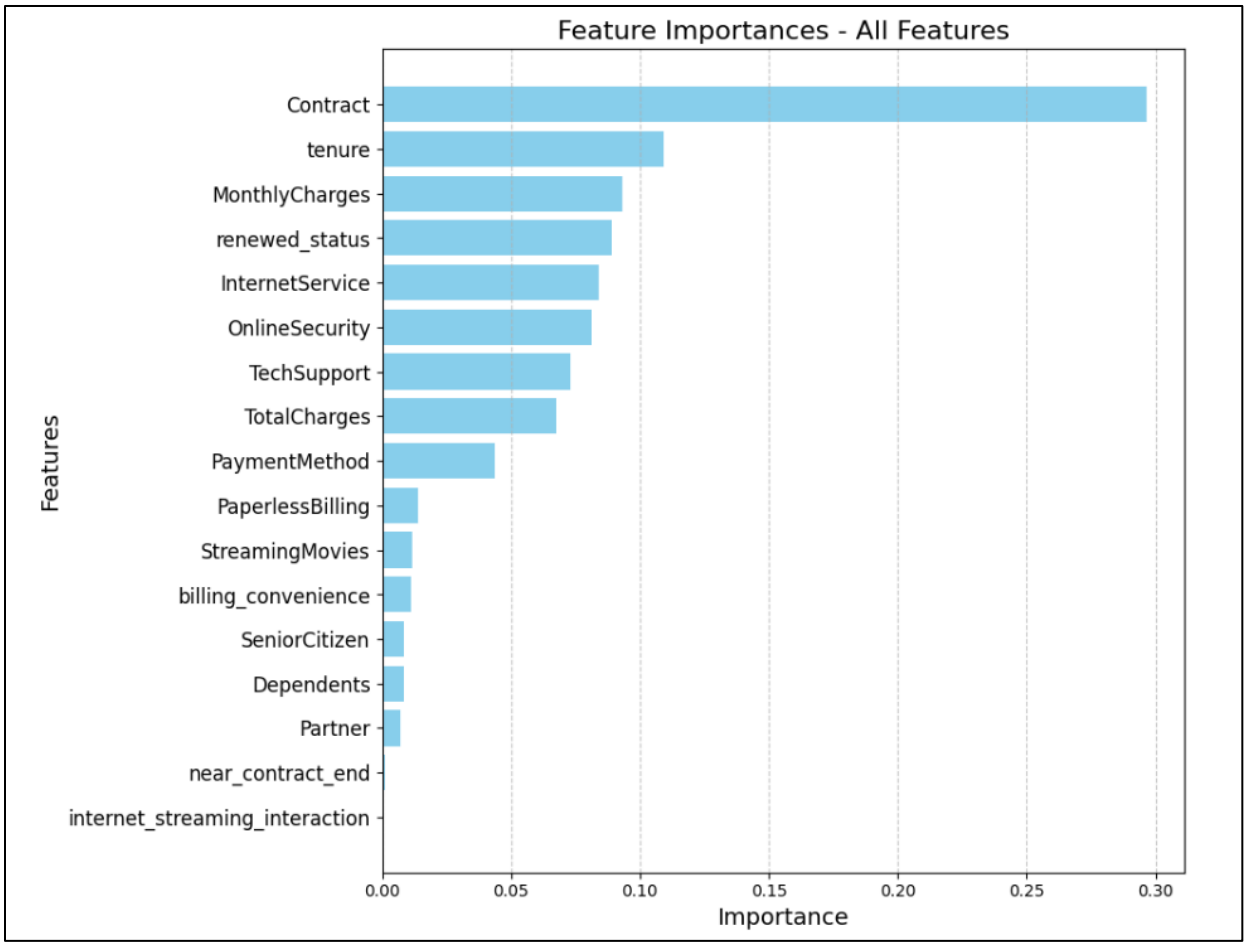


Figure 21. Bar chart of feature importance

```
Number of features: 1, Best recall Score: 0.7772183363538681, Best Params: {'n_estimators': 50, 'min_samples_split': 10, 'min_samples_leaf': 1, 'max_depth': 10, 'bootstrap': True}
Number of features: 2, Best recall Score: 0.7789135443901496, Best Params: {'n_estimators': 50, 'min_samples_split': 10, 'min_samples_leaf': 1, 'max_depth': 10, 'bootstrap': True}
Number of features: 3, Best recall Score: 0.8289845440358834, Best Params: {'n_estimators': 100, 'min_samples_split': 2, 'min_samples_leaf': 4, 'max_depth': 10, 'bootstrap': True}
Number of features: 4, Best recall Score: 0.8422897312553616, Best Params: {'n_estimators': 200, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_depth': 10, 'bootstrap': False}
Number of features: 5, Best recall Score: 0.8427739927565723, Best Params: {'n_estimators': 100, 'min_samples_split': 2, 'min_samples_leaf': 4, 'max_depth': 10, 'bootstrap': True}
Number of features: 6, Best recall Score: 0.8330957895014215, Best Params: {'n_estimators': 50, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_depth': 10, 'bootstrap': True}
Number of features: 7, Best recall Score: 0.8522039168381882, Best Params: {'n_estimators': 200, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_depth': 10, 'bootstrap': False}
Number of features: 8, Best recall Score: 0.8592198529648574, Best Params: {'n_estimators': 200, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_depth': 10, 'bootstrap': False}
Number of features: 9, Best recall Score: 0.8630910171541, Best Params: {'n_estimators': 200, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_depth': 10, 'bootstrap': False}
Number of features: 10, Best recall Score: 0.8635744003091779, Best Params: {'n_estimators': 200, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_depth': 10, 'bootstrap': False}
Number of features: 11, Best recall Score: 0.8611566061876557, Best Params: {'n_estimators': 200, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_depth': 10, 'bootstrap': False}
Number of features: 12, Best recall Score: 0.8633334406867496, Best Params: {'n_estimators': 200, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_depth': 10, 'bootstrap': False}
Number of features: 13, Best recall Score: 0.8630907243720557, Best Params: {'n_estimators': 200, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_depth': 10, 'bootstrap': False}
Number of features: 14, Best recall Score: 0.8645452655679533, Best Params: {'n_estimators': 300, 'min_samples_split': 10, 'min_samples_leaf': 4, 'max_depth': None, 'bootstrap': False}
Number of features: 15, Best recall Score: 0.8652687299993266, Best Params: {'n_estimators': 200, 'min_samples_split': 10, 'min_samples_leaf': 4, 'max_depth': 40, 'bootstrap': False}
Number of features: 16, Best recall Score: 0.8635746930912221, Best Params: {'n_estimators': 200, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_depth': 10, 'bootstrap': False}
Number of features: 17, Best recall Score: 0.8679307043457639, Best Params: {'n_estimators': 200, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_depth': 10, 'bootstrap': False}

Best number of features: 17
Best hyperparameters: {'n_estimators': 200, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_depth': 10, 'bootstrap': False}
Best cross-validation {scoring_method} score: 0.8679307043457639
```

Figure 22. Number of features after RFE

6. Model Building

Before building the model, we split the data into train and test sets. Eighty percent of the data is a train set, and the rest of the data is a test set.

6.1 Stacking

In a Stacking model, Random Forest and Gradient Boosting are used as base learners and the final output is predicted by a Logistic Regression model. This process is done in a way of stacking to improve the performance of the model.

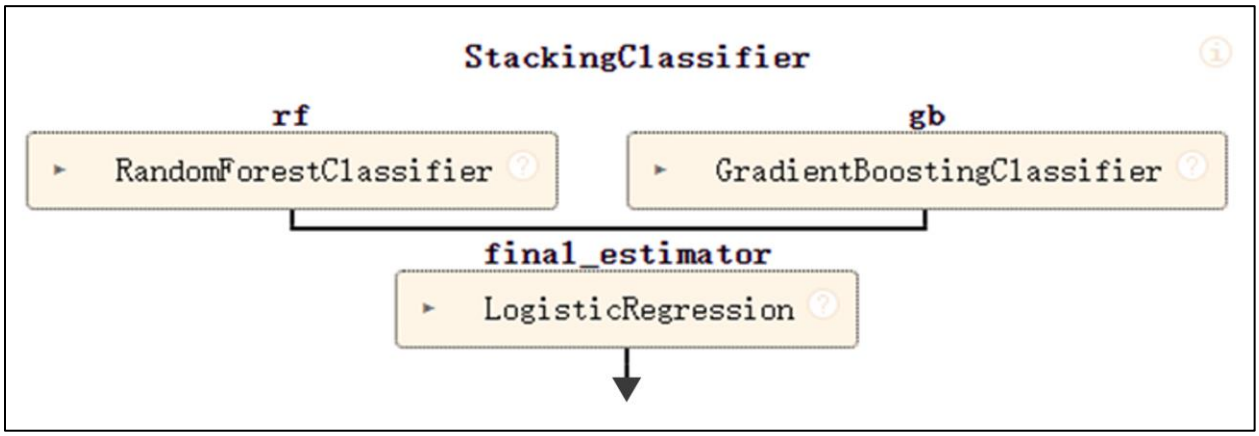


Figure 23. Principles of the Stacking Model

6.2 Modeling Process

Firstly, random forest is chosen as the basic classifier to optimize the feature set and hyper-parameter settings of the machine learning model through Recursive Feature Elimination and Randomized Search Cross-Validation, which helps to improve the performance of the model by efficiently finding the optimal features and parameters through an iterative approach.

After that we are able to obtain a ranking of the importance of all features for the model and eliminate those that are not important for the model, then fit the data with the features that are important to the model, and bring the new data into the Stacking model for training and prediction.

7. Model Evaluation

7.1 The metrics for evaluation

To evaluate the accuracy of the model's predictions, we output the Confusion Matrix and Classification Report.

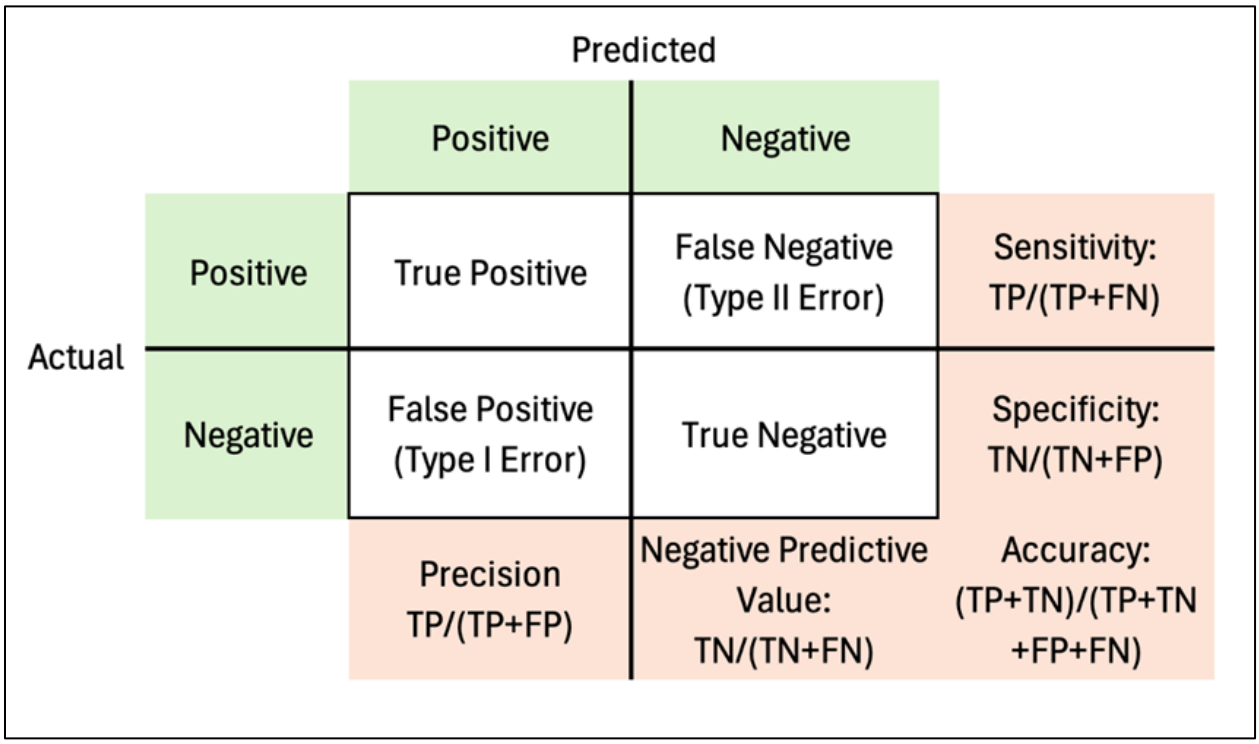


Figure 24. Contents of Confusion Matrix and Classification Report

7.2 The Output of Stacking Model

We can see from the output of the Stacking model that the number of people who were correctly predicted as No and Yes were 843 and 877 respectively, and the accuracy metrics of the model's predictions, precision, and recall, were both significantly higher than 0.8.

```
Confusion matrix:
[[843 194]
 [152 877]]
Classification Report:
              precision    recall  f1-score   support

         0.0       0.85      0.81      0.83      1037
         1.0       0.82      0.85      0.84      1029

    accuracy                           0.83      2066
   macro avg       0.83      0.83      0.83      2066
weighted avg       0.83      0.83      0.83      2066

Accuracy:
0.8325266214908035
```

Figure 25. Outputs of Stacking

We also use the RF model for prediction, we can see that the random forest model's ability to accurately predict is weaker than the stacking model.

```
Confusion matrix:
[[811 226]
 [143 886]]
Classification Report:
              precision    recall  f1-score   support

         0.0       0.85      0.78      0.81      1037
         1.0       0.80      0.86      0.83      1029

    accuracy                           0.82      2066
   macro avg       0.82      0.82      0.82      2066
weighted avg       0.82      0.82      0.82      2066

Accuracy:
0.8213939980638916
```

Figure 26. Outputs of RF

8. Key Factors and Strategy

The predictive churn analysis identifies **419 customers** who are likely to churn. We will utilize the Feature Importance graph, generated by our prediction model, to gain a clear understanding of the key factors influencing customer churn.

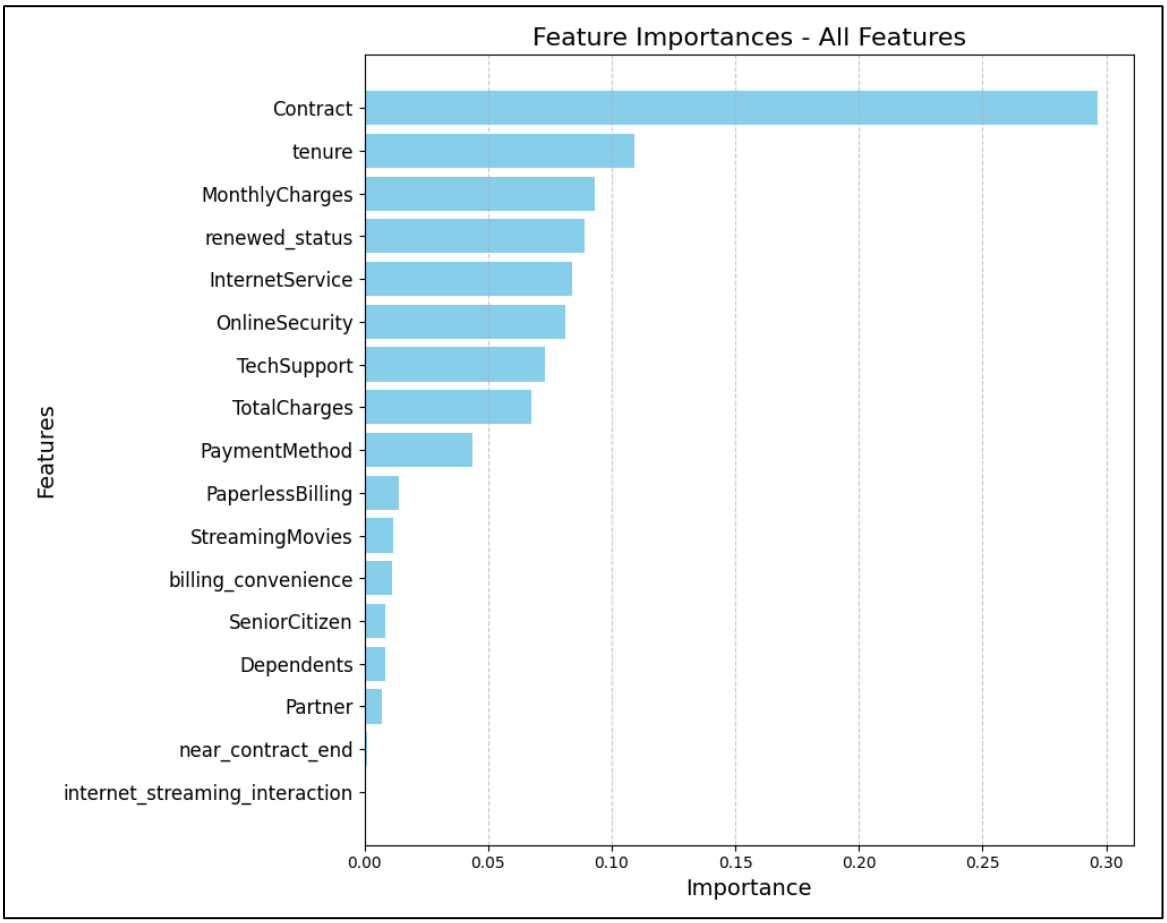


Figure 27. Feature importance

8.1 Contract Type & Renewed Status:

Contract type has the greatest impact on churn, with single-month contract users at the highest risk of churn.

For single-month contract users, introduce them to flexible contract options such as 3 or 6-month, etc., to attract users who are not sure of their long-term binding contracts. In the expiring period and early renewal period (1-7 months), proactively contact the subscriber and offer renewal discounts or service upgrade options to enhance retention.

8.2 Tenure:

Users with a tenure of 0-10 months are at higher risk of churn, offer new users a discount on the first month or additional free services (e.g. online security) to increase the attractiveness of the initial experience, improve new user satisfaction, and reduce early churn.

8.3 Monthly Charges:

According to the Global Telecommunications Study 2022, after the epidemic more and more users found telecoms expensive, rising from 51% to 58%. (2)

Subscribers with monthly charges in the 60-110 range are at higher risk of churn, usually because they expect higher quality of service.

For DSL subscribers with high monthly charges, offer fibre-optic network upgrades with first-month discounts or free upgrades.

8.4 Internet Service, Online Security & Tech Support

According to Oliver Wyman's analysis, The most important considerations for EU consumers in choosing a teleco are price, network services and data allowance. (3)

Fibre optic subscribers have high expectations of network speed and stability and are prone to churn once they are dissatisfied. And those who have not subscribed to Online Security or Tech Support are at high risk of churn.

Provide fibre-optic users with network quality monitoring and regular feedback, with value-added services, to ensure that their high service needs are met with timely resolution of any customer complaints. For potential customers, provide free Online Security or Tech Support services.

9. Conclusion

By using targeted strategies among potential churn users, Telco can significantly reduce churn rates, improve customer satisfaction, and enhance profitability.

Reference List

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(2) Villegas, V. (2022) *How telcos can keep their customer base amidst a high churning rate*, *Asian Business Review*. Available at: https://asianbusinessreview.com/exclusive/how-telcos-can-keep-their-customer-base-amidst-high-churning-rate (Accessed: 27 October 2024).

(3) Amiot, E., Arriba, P. de, Youssef, J., Willenbruch, H., Raffaele, E., Lacave, B. and Bosch, L. M. del (2023) *Analyzing Churn Challenges In The European Telecom Landscape, Oliver Wyman - Impact-Driven Strategy Advisors*. Available at: https://www.oliverwyman.com/our-expertise/insights/2023/oct/european-telecoms-churn-trends.html (Accessed: 27 October 2024).

Appendix

Appendix 1 - [Data Engineering] Link to Github with code:
https://github.com/qweqwedfa/DataAnalyticsGroup1/tree/main/DE

Appendix 2 - [Data Visualization] Link to Github with code:
https://github.com/qweqwedfa/DataAnalyticsGroup1/tree/main/DV

Appendix 3 - [Data Science] Link to Github with code:
https://github.com/qweqwedfa/DataAnalyticsGroup1/tree/main/DS

Appendix 4 - [Prediction Result Visualization] Link to Github with code:
https://github.com/qweqwedfa/DataAnalyticsGroup1/tree/main/Prediction_Visualization